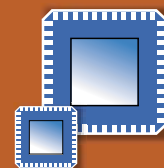




EXPERIENCE **REAL**
WIRELESS CONNECTIVITY

Wisair 531



Wisair 531 Integrated MAC\Baseband UWB chip

Pre-production 2nd Generation UWB WiMedia\MBOA-Compliant Chip

General Description

The Wisair 531 is a 2nd generation CMOS-based MAC\Baseband chip, part of Wisair's Ultra Wideband (UWB) chipset solution.

Integrated into Wisair's Development Kit (DVK) and reference designs, the Wisair 531 is WiMedia\MBOA compliant, enabling the rapid development of UWB-based consumer electronics and wireless USB applications running on mobile devices, laptops, PC's, and PC peripherals.

Highlights

- Second generation, pre-production chipset
- WiMedia\MBOA-compliant
- Supports WUSB and Wireless Ethernet/IP reference designs
- Link robustness to channel variance
- Subset (proprietary) of MBOA TDMA-based MAC
- Supports external MAC using MBOA MAC-PHY Interface

Key Features

General

- Data rates up to 480Mbps
- Frequency range from 3.1 to 4.8 GHz
- Range supported >10 meters
- Power consumption less than 300mW
- Robust wireless link communication
- Supports FFI and TFI sequence schemes

- Supports TPC with 16 transmit power levels
- Operation temperature range -40 oC to +85oC
- Based on a CMOS 0.13u technology
- TFBGA 225 balls, 13 x 13 mm package
- Coexistence with 802.11a/b/g with no external filtering

Integrated on-chip

- On-chip A/D and D/A high speed converters for I & Q channels
- On-chip memory (no external memory needed)

Compliance

- WiMedia\MBOA PHY specification Rev 1.0
- WiMedia\MBOA MAC-PHY interface specification Rev0.99
- FCC Part 15 Class B

Interfaces

- Data Interfaces:
 - RMII, MII (MII-PHY and MII-MAC modes available) using Full-duplex internal basic Ethernet 802.3 MAC
 - Direct CPU Interface (DCI): SRAM-like interfaces with 3 available operational modes: Synchronous 8 bit and Asynchronous 8/16 bit
 - MPEG TS SPI (Synchronous Parallel Interface) to connect to transport stream applications (SPI-Rx and SPI-Tx modes)
- Host control Interfaces:
 - Intel-like parallel 8 bit asynchronous multiplexed bus
 - Configurable through XPLME serial interface (MAC-PHY interface)
- WiMedia\MBOA MAC-PHY interface

Reference Design

Wisair reference designs target wireless USB applications and fast Ethernet/IP communication. Using the Wisair chipset in an optimal way to minimize the Bill-of-Material (BOM), size and power consumption, reference designs include the 531 MAC/Baseband chip, the 502 transceiver IC and a differential antenna connector. Different antennas are available as part of the reference design.

External supporting components include a single off-the-shelf crystal and passives. The assembled 531 and 502 reference design circuitry covers a small footprint on the board's space of 21 by 32 mm. Also included is a software API and WisMan™ Windows-based reference application for controlling and managing the chipset.

Wisair Multiband OFDM UWB chipset

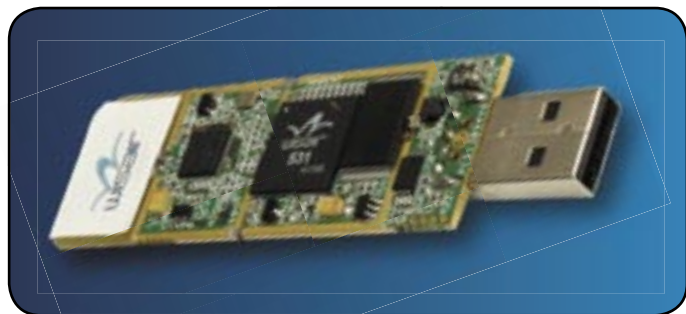
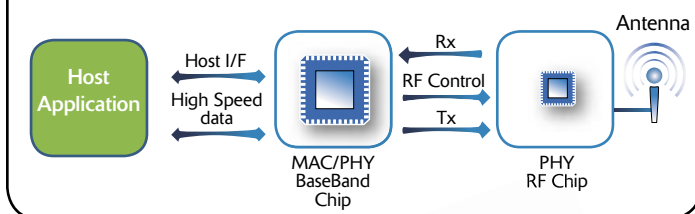
Wisair offers a comprehensive solution based on a UWB chipset, software API, and a reference design systems for demonstration, evaluation and product development purposes.

The Wisair UWB chipset provides a high bit rate, low power consumption and low cost solution for Wireless Personal Area Network (WPAN) applications. It supports short-range home/office wireless connectivity, multi-streaming of high quality video, fast upload/download of content and broadband wireless multi-media sharing.

Wisair's chipset consists of:

- Wisair 502 PHY RF chip
- Wisair 531 MAC\ Baseband chipset

Wisair Chipset



480 Mbps Ultra Wideband USB host dongle
reference design

About Wisair

Wisair develops and markets Ultra Wideband, high performance wireless chipset solutions. The company is an active promoter member of the WiMedia Alliance, an active participant in the IEEE standardization process, and a member of the Wireless USB promoter group.

A privately-held company, and part of the RAD group, Wisair is headquartered in Israel with offices in USA, Japan, Korea and Taiwan.